

AI TRAVEL PLANNER

Professional Project Report

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1. Project Overview

AI Travel Planner is an intelligent travel planning web application developed using Python and Streamlit. The system helps users generate customized travel itineraries based on destination city, budget, and number of travel days. The platform provides hotel recommendations, weather forecasts, budget estimation, tourist attractions, and downloadable PDF reports.

2. Objectives

- Build a smart AI-based travel planning assistant.
- Provide luxury and budget-friendly trip recommendations.
- Display weather forecasts and travel tips.
- Generate attractive travel itineraries.
- Deploy the project online using Streamlit Cloud.

3. Technologies Used

Frontend: Streamlit

Backend: Python

Libraries: FPDF, JSON, Requests

Version Control: Git & GitHub

IDE Used: Visual Studio Code

4. System Architecture

The application follows a modular architecture where frontend UI components communicate with backend logic modules. The backend processes user inputs and fetches relevant hotel, weather, and tourist attraction data from local JSON datasets and APIs.

Architecture Flow:

- User Input → Streamlit UI
- UI Sends Data → Main Python Logic
- Travel Tools Process Data
- Hotels / Weather / Places Retrieved
- Travel Itinerary Generated
- PDF Export Created

5. Features & Functionalities

- Luxury UI Design
- Destination Selection Dropdowns
- Smart Budget Calculation
- Weather Forecast Display
- Hotel Recommendations
- Tourist Attractions Suggestions
- PDF Travel Report Download
- Responsive Streamlit Dashboard

6. How I Built This Project

Step 1: Designed the project structure and folders.

Step 2: Created tools for hotels, places, weather, and budget calculation.

Step 3: Built the backend logic in main.py.

Step 4: Designed a premium Streamlit UI.

Step 5: Added PDF export functionality.

Step 6: Tested and debugged the application.

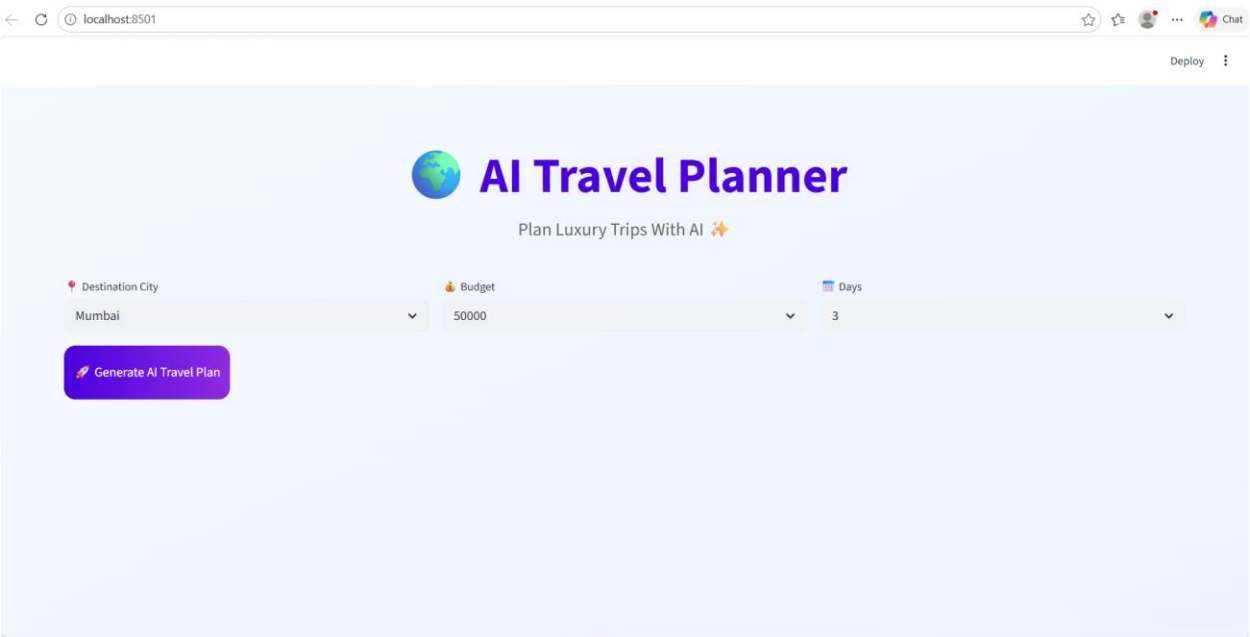
Step 7: Prepared GitHub repository and deployment setup.

7. Implementation Workflow

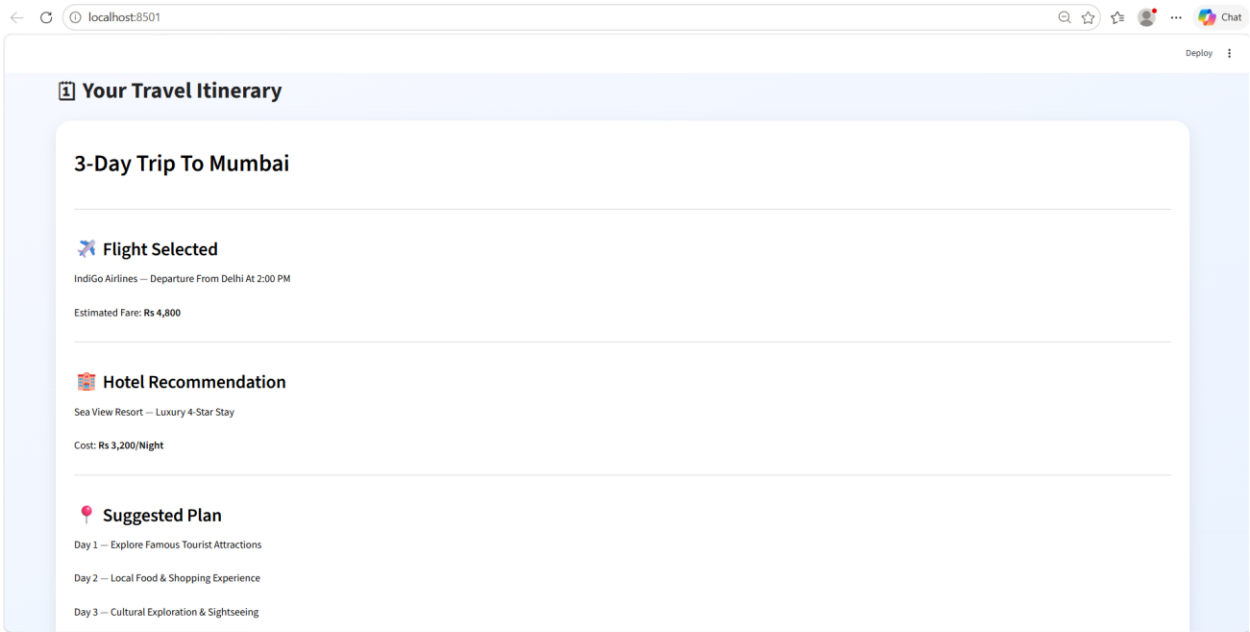
1. User selects city, budget, and trip duration.
2. Backend processes travel requirements.
3. Hotels and tourist attractions are recommended.
4. Weather forecast is displayed.
5. Budget breakdown is calculated.
6. Final travel itinerary is generated.
7. User downloads the PDF report.

8. UI & Screenshots

Main Dashboard UI



Travel Itinerary & Recommendations



Weather Forecast & Budget Breakdown

Weather Forecast

 Day 1: Sunny — 31°C

 Day 2: Partly Cloudy — 29°C

 Day 3: Pleasant Breeze — 28°C

Budget Breakdown

 Flights: Rs 4,800

 Hotels: Rs 6,400

 Food & Local Travel: Rs 2,500

Total Estimated Cost: Rs 13,700

9. GitHub Repository

GitHub Repository:

<https://github.com/mohittaluja1111/Agentic-AI-Travel-Planner>

10. Conclusion & Future Enhancements

The AI Travel Planner successfully demonstrates how AI-powered systems can simplify travel planning. The project provides an attractive and interactive user experience while generating personalized travel itineraries.

Future Enhancements:

- Real-time flight APIs
- AI chatbot integration
- Voice-based travel planning
- Multi-language support
- Hotel image previews and maps integration

THANK YOU